

Project Gutenberg language diversity over time

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Abstract

languages, with English comprising 81% of all books. Their work established baseline metrics for understanding linguistic representation in digital libraries.

1 Introduction

This study analyses the linguistic diversity in Project Gutenberg's digital library collection over time, examining changes in the additions for each language in the Gutenberg Collection per year. The research aims to understand how increased internet access in non-English speaking countries may have influenced digital preservation efforts. It does this by analysing the works added year over year to the Gutenberg collection, showing the changes in addition rates of differing languages.

2 Related Work

Several studies have examined linguistic diversity in digital libraries and Project Gutenberg specifically:

1. Whitt (2018) explored multilingualism on the web and Project Gutenberg's early efforts at linguistic diversification, providing important historical context for understanding the evolution of digital preservation efforts.
2. Gerlach and Font-Clos (2020) analysed a curated Project Gutenberg database, revealing a strong bias toward Western lan-

3 Data

Pre-processing The data pre-processing was done by downloading the metadata of all Project Gutenberg books from the [gutenberg.org](https://www.gutenberg.org) website on 06-01-2025. This data can be downloaded in the form of a CSV file, which contains the relevant fields for this article: the date added and the language. Books that included multiple languages, either due to being a translation that includes the original text or being a translation dictionary, are counted for both languages. The python script used for extracting the data will be included with the paper. Due to not being completed, and thus giving incomplete information, the year 2025 was excluded. As only the language codes were given with the metadata, language codes were manually linked to their respective language. This resulted in a matrix for book additions per year for every language, from the years 1971 to 2024, with 70 languages. Trying to process this data directly

4 Predicted Results

We expect that the data will show an increase in the proportion of non-English books in Project Gutenberg's collection over time. This is based on the idea that increased internet access in non-English speaking countries leads to greater easier access and participation with Project Gutenberg. Additionally, we anticipate that the diversity of languages represented will have broadened, reflecting a more globalized effort in digital preservation.

Discussion

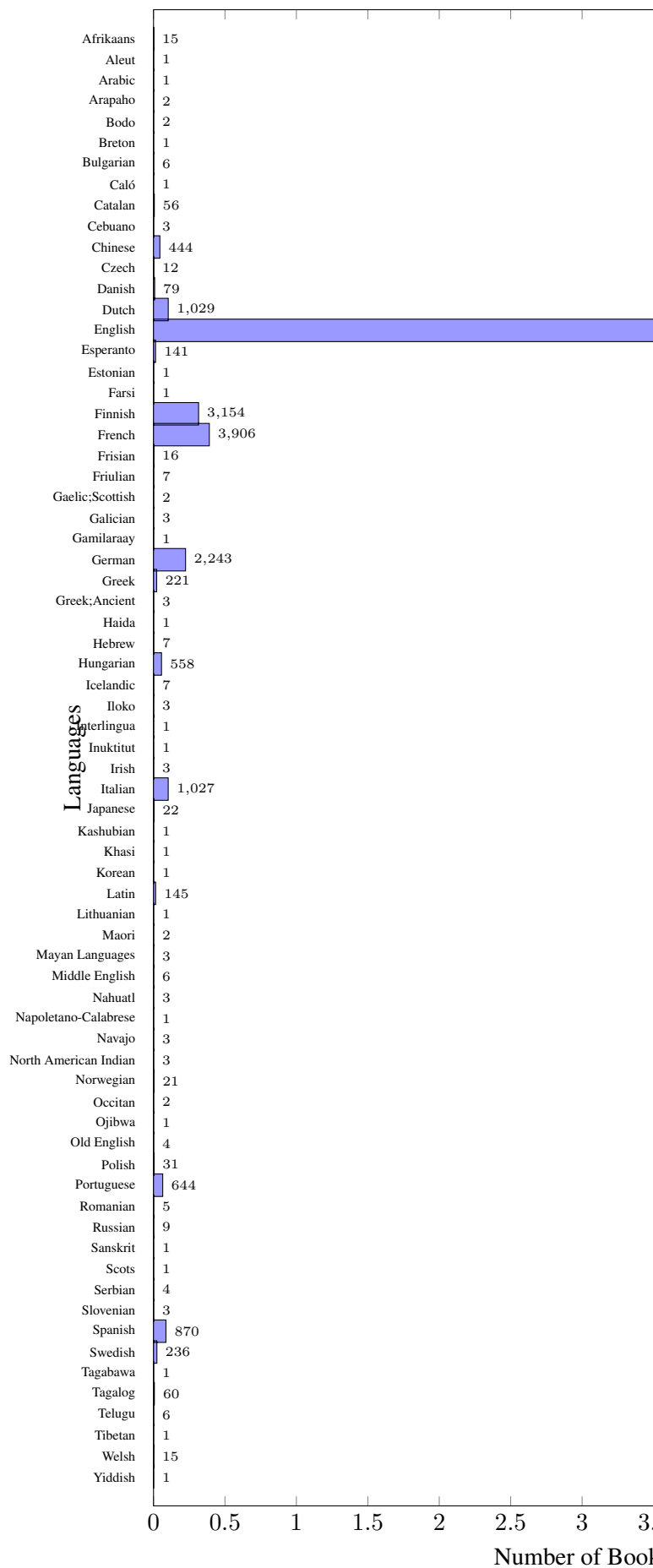
5 Conclusion

References

- Gerlach, Martin and Francesc Font-Clos (Jan. 2020). "A Standardized Project Gutenberg Corpus for Statistical Analysis of Natural Language and Quantitative Linguistics". In: *Entropy* 22.1. Number: 1 Publisher: Multidisciplinary Digital Publishing Institute, p. 126. ISSN: 1099-4300. DOI: 10.3390/e22010126. URL: <https://www.mdpi.com/1099-4300/22/1/126> (visited on 12/12/2024).
- Whitt, Richard J. (2018). *Diachronic Corpora, Genre, and Language Change*. Amsterdam/Philadelphia, NETHERLANDS, THE: John Benjamins Publishing Company. ISBN: 978-90-272-6350-6. URL: <http://ebookcentral.proquest.com/lib/rug/detail.action?docID=6733930> (visited on 12/05/2024).

6 Appendix

6.1 Appendix A



[illegible]

Table 1: The number of books added to Project Gutenberg per year per language